Understanding Saudi Road Freight Operations

A practical study guide for building freight software in Saudi Arabia

A freight operator sells a promise: collect particular goods, deliver them under agreed conditions, and provide the evidence needed to charge for the work. Keeping that promise requires coordinating vehicles, drivers, cargo, facilities, documents and money.

For the dispatch product we are considering, the central question is:

> Given the work we have promised, which available resources should do it, in what order, and what should we change when reality disrupts the plan?

This guide starts with the wider logistics industry, then concentrates on domestic road freight and the work of a carrier's dispatch team. Read Sections 1–6 first, work through Sections 8–9 with a pen, and use the glossary and study plan afterward. The numerical examples are fictional teaching cases, not Saudi market rates or legally approved operating schedules. Saudi service references were checked on 10 October 2026; live operating requirements must come from the applicable current rules and contracts.

1 How logistics fits together

Supply chain management covers how a business sources, makes and supplies products. **Logistics** covers the movement, storage and handling needed along that chain. **Transport** is the movement between locations.

Imagine a Saudi distributor buying appliances from an overseas manufacturer. The goods may travel from the factory to an export port, cross the sea, clear customs, reach a Saudi warehouse by truck, and later go to retailers. A sale, a customs clearance and a truck trip are different activities, even when they concern the same goods.

Our current focus is the company operating those truck movements. It might carry its customers' goods between cities, replenish warehouses or move freight from a port. Its dispatch team decides how to deploy capacity.

Three flows run through every transport operation:

Flow	What moves	Example
Physical	Goods, vehicles and drivers	A loaded trailer travels from a warehouse to a customer
Information	Requests, commitments, instructions and evidence	An order gives a delivery window; a driver reports arrival

Flow	What moves	Example
Financial	Prices, charges, costs, invoices and payments	The carrier pays for fuel, invoices the customer and later receives payment

These flows do not finish together. Goods may be delivered while the proof of delivery is missing. The invoice may be issued while the customer disputes a waiting charge. A truck may complete its work before a subcontractor sends its bill.

As you learn, repeatedly ask: **What happened physically, what do we know about it, and what does it mean financially?**

2 Who is involved

The company paying for transport is not always the company sending or receiving the goods.

Party	Its role	What matters to it
Shipper or consignor	Sends the goods	Collection, correct handling and a reliable handover
Consignee or receiver	Receives the goods	Delivery timing, quantity, condition and receiving capacity
Freight payer	Pays the transport invoice under the agreement	Correct rates, approved charges and supporting documents
Carrier or haulier	Undertakes the road transport	Feasible work, reliable execution and profitable use of capacity
Subcontractor	Performs work assigned by another carrier or logistics provider	Clear instructions, agreed pay and prompt settlement
Freight broker or transport intermediary	Arranges capacity between customers and carriers	Carrier availability, commercial terms and service accountability
Freight forwarder	Organizes transport chains and related services, often across modes and borders	Coordination between carriers, documentation and trade requirements
Third party logistics provider or 3PL	Performs outsourced logistics, potentially including warehousing and transport	Performance of the broader customer operation
Customs broker	Handles authorized customs declaration and clearance work	Accurate trade documents and release of goods

These are functional descriptions. One business can perform several roles; its legal responsibilities depend on its actual contracts and licences. FIATA's explanation of freight forwarding is useful for understanding why a forwarder can coordinate a transport chain without physically operating every leg.

Different kinds of fleet businesses

An **own-fleet carrier** operates vehicles it owns or leases. A **mixed-fleet carrier** uses its controlled vehicles and purchases additional capacity from partners. An **asset-light operator** relies primarily on contracted transport. A **private fleet** moves its parent company's own products.

For planning software, ownership alone is not the deciding factor. The useful question is **who can assign the vehicle, and for what period?**

A leased truck might be fully controlled by the dispatcher. A truck owned by the company might be committed to a dedicated customer for the entire week. A partner's quotation might offer a service without reserving any capacity.

People inside the carrier

Sales agrees what the customer can buy. Order entry turns the request into an operational job. Dispatch assigns and sequences the work. Drivers execute it and report events. Fleet maintenance determines vehicle availability. Customer service communicates changes. Finance checks charges and documents, invoices customers and pays suppliers.

In a small business, one person may cover several of these jobs. The handovers still exist, even when they happen inside that person's head.

3 Freight segments have different workflows

“Logistics software” is too broad a category to understand a customer's daily work. First identify the operating pattern.

Segment	How the freight moves	Main planning difficulty
Full truckload or FTL	A vehicle or trailer is dedicated to a customer's load	Matching equipment and drivers, sequencing trips and positioning for later work
Less than truckload or LTL	Several customers share capacity	Consolidation, collection and delivery routes, transfer timing and capacity
Local distribution	One vehicle makes multiple deliveries or collections	Stop order, access, unloading time and customer windows
Container haulage or drayage	Containers move between ports, depots and inland sites	Releases, appointments, chassis, container identity and empty returns
Refrigerated transport	Cargo travels under specified temperature conditions	Equipment, temperature evidence, compatibility and handling
Bulk or tanker transport	Loose material or liquid moves in specialist equipment	Payload, product compatibility, cleaning and unloading facilities
Heavy or abnormal loads	Oversized or unusually heavy cargo moves on specialist equipment	Route suitability, permits, escorts and engineering constraints

FTL does **not** mean every cubic metre is occupied. A customer may reserve the entire vehicle for urgency, security or handling reasons. LTL does not always mean a terminal network; some operators combine part loads directly. Learn the particular company's service design. DHL's road-freight services illustrate the distinction between groupage networks, direct transport and specialist services.

Why LTL becomes a different planning problem

A two-pallet shipment could be collected by a local truck, transferred to an origin terminal, carried with other customers' goods on an intercity truck, and delivered by another local truck.

The customer bought one transport service. Internally, the operator performed several **legs**. Many shipments can share one leg, and one shipment can use several vehicles.

A **cross-dock** transfers freight between incoming and outgoing transport with limited storage. A **warehouse** primarily stores and manages inventory. Both can exist at the same site, but their operational purposes differ.

The initial product in our PRD concentrates on dedicated jobs with one pickup and one delivery, followed by other jobs on the same resources. That gives you a clear first workflow to master before adding consolidation and terminal connections.

4 The things dispatchers actually schedule

A truck can represent several separate resources

A **rigid truck** has its cargo body attached to the vehicle. An articulated combination usually includes a **tractor unit**, a **semi-trailer** and a **driver**.

The tractor might be mechanically available while its trailer is still being unloaded. The trailer might be ready while the driver is resting. A replacement driver cannot take over unless they can physically reach the vehicle and are eligible to operate it.

A fleet count of 100 vehicles therefore does not mean 100 interchangeable resources available now.

For each resource, ask:

- Where is it, and how recently was that location confirmed?
- What is it doing, and when is it expected to be released?
- What cargo, equipment or customer restrictions apply?
- What future work is already committed?
- What maintenance, working-time or access constraints apply?

Availability has a location, a time and a set of conditions. It is rarely a simple yes or no.

Cargo uses several kinds of capacity

Weight, volume, floor space and handling requirements are separate constraints.

Dense cargo can exhaust the permitted payload while leaving room in the trailer. Light bulky cargo can fill the body while using little payload. Non-stackable pallets consume more usable floor space than their volume suggests. Cargo may fit geometrically but have unacceptable weight distribution or require loading equipment the site lacks.

Useful units include kilograms or tonnes, cubic metres, pallet positions and **loading metres**, a measure of occupied trailer floor length. A pallet is a handling unit, not a universal size or weight. Actual dimensions and stackability matter.

Locations are operating sites

"Riyadh" is not a sufficient pickup address. A job may need a particular warehouse gate, entry route, contact, appointment, loading bay and security approval.

A site's opening hours differ from an appointment. "We receive from 08:00 to 17:00" does not mean every truck can arrive at any moment and be unloaded immediately.

Lane and network

A **lane** is an origin-to-destination flow, such as Dammam to Riyadh. The reverse direction is a separate commercial market: demand and available trucks can differ.

A **depot** is an operating base. A **hub** concentrates flows for onward movement. **Linehaul** generally refers to the longer movement between hubs, terminals or major locations.

Where a truck ends today affects the work it can perform tomorrow. This is why planning one shipment independently can produce a poor fleet plan.

5 One shipment from request to payment

Consider a fictional distributor asking a carrier to move a dedicated load of packaged goods from a Dammam warehouse to a Riyadh warehouse.

Step 1 Understand the request

The customer supplies some combination of locations, cargo details, collection date and delivery requirement. Requests can arrive through a portal, system integration, spreadsheet, email or a conversation.

The order-entry person must establish what is missing. Does "tomorrow morning" refer to pickup or delivery? Is the cargo ready? Does the destination unload the vehicle? Is the weight total cargo weight or weight per pallet? Is a particular trailer required?

A record can exist before it is safe to plan. "Captured" and "operationally ready" should mean different things.

Step 2 Agree price and service

The carrier uses a negotiated rate agreement or prepares a spot quote for this request. The commercial agreement should establish the base service, what is included, extra charges and payment terms.

A quote does not necessarily become a confirmed job. Acceptance, capacity confirmation and any required customer authorization are separate events.

A transport **tender** can mean a request to a carrier to accept an individual shipment. In another context it means bidding for a longer contract. Always establish which meaning is intended.

Step 3 Check the job before planning

Operations confirms addresses, time windows, equipment, cargo readiness and relevant documents. Credit control may have a separate role if the customer is on hold.

A pickup window might mean the time to arrive at the gate, start loading or finish loading. These are different constraints. A usable order states which event the customer expects.

Step 4 Assign resources and sequence the work

The dispatcher chooses the vehicle, trailer and driver, or books an approved subcontractor. They consider the empty journey to pickup, loading, travel, unloading, driver availability and the next commitment.

The nearest truck may be wrong because it is unsuitable, still busy or needed for work that other trucks cannot do. The dispatcher is coordinating the day's work as a whole.

Step 5 Publish and acknowledge instructions

A plan in the office is a proposal until the execution team receives it. Drivers need the correct address, contact, times, cargo instructions and required references.

A sent instruction is not proof that the driver has understood and accepted it. Changes must reach the right person; superseded instructions must not remain ambiguous.

Step 6 Collect the goods

The truck travels to the site, checks in, waits if necessary, loads and departs. The driver or loading team records the relevant quantity, condition, references and exceptions.

Keep arrival, loading start, loading completion and departure distinct. A two-hour delay could come from a gate queue, unavailable cargo or loading itself. Those causes suggest different improvements.

Step 7 Monitor and respond

Travel may encounter traffic, access restrictions, breakdowns or changes at the destination. The dispatcher updates the expected arrival and checks downstream work.

An **ETA**, or estimated time of arrival, is a prediction. It is not the same as the promised delivery window or the actual arrival event.

Step 8 Deliver and establish the outcome

The destination receives the goods, checks quantity and condition, and records acceptance or exceptions. A partial delivery, damage or refusal requires a different follow-up from a clean delivery.

Proof of delivery, or POD, may include a signed document, electronic signature, quantities, timestamps, photographs or a receiver reference. The customer's agreed evidence requirements determine what is sufficient.

Step 9 Prepare the commercial closeout

Finance checks the agreed price, additional services, evidence of waiting or other charges, and required references. The carrier may also need to match a subcontractor's bill to the performed work.

Delivered work can remain unbilled because evidence or approval is missing. Conversely, some commercial arrangements bill in advance or periodically. The example here follows a common post-delivery workflow, not a universal invoicing rule.

Step 10 Invoice and collect

The transport invoice enters the customer's approval and payment process. Disputes, credits, partial payments and reconciliation can follow.

Operational completion, invoice readiness, invoice issuance and payment are separate milestones. Opter's order-status documentation illustrates this distinction by separating dispatch readiness, invoicing readiness and execution status.

```
flowchart TD
    A["Customer request"] --> B{"Details and commitment complete"}
    B -->|No| C["Resolve missing information"]
    C --> B
    B -->|Yes| D["Assign resources and publish"]
    D --> E["Collect and transport"]
    E --> F{"Can the plan still be fulfilled"}
    F -->|No| G["Confirm recovery and notify"]
    G --> E
    F -->|Yes| H["Deliver and record outcome"]
    H --> I{"Billing evidence complete"}
    I -->|No| J["Resolve evidence or dispute"]
    J --> I
    I -->|Yes| K["Invoice and collect"]
```

6 What a dispatcher does during the day

The dispatcher manages a changing set of commitments. Routing is one part of that job.

At the start of a shift, they reconcile overnight events: which trips finished, which trucks remain at customers, which drivers are available, what broke down and what new orders arrived. They then cover imminent pickups and identify the jobs that have few feasible options.

As the shift progresses, the dispatcher deals with new information. A customer postpones collection. A vehicle becomes free earlier than expected. A partner fails to confirm. A delivery takes longer than planned. Each event can change several later assignments.

Toward the end of the shift, the team prepares the next handover: outstanding work, uncertain release times, instructions already sent and customers needing attention.

Four distinctions help you understand the work:

Distinction	Why it matters
Hard constraint versus preference	A vehicle cannot exceed applicable limits; a preference for a particular driver may be negotiable
Confirmed versus tentative	A booked partner truck can support a plan; an unanswered enquiry cannot
Planned versus actual	The schedule says unloading ends at 14:00; a confirmed departure tells you what happened
Local versus wider consequence	Saving a short empty movement now may prevent a much better assignment later

When a dispatcher overrides software, investigate the reason. They may know a constraint that the system lacks, distrust stale data, protect a customer commitment or make an avoidable habit-based choice. The override itself does not tell you which explanation is correct.

7 The documents and what they prove

Documents connect physical events to commercial and regulatory responsibilities. Similar names can hide very different purposes.

Document or record	What it describes	What it does not establish by itself
Customer transport request	The service the customer wants	That the carrier accepted it
Quote or rate agreement	Commercial terms and service scope	That a vehicle is reserved
Purchase order or PO	Customer authorization to purchase	That the goods have been delivered
Goods commercial invoice	The sale and value of the goods	The carrier's transport charge

Document or record	What it describes	What it does not establish by itself
Packing list	Packages, contents, quantities and relevant measurements	A confirmed vehicle assignment
Transport order or job	The operator's record of work to perform	That dispatch instructions reached the driver
Road consignment note or transport document	Parties, goods and details of the carriage	That payment has been received
Manifest or load list	Goods or shipments included in a movement	Acceptance of every item at destination
Delivery note	Goods presented for delivery	Accepted delivery unless the required acknowledgement is recorded
POD	Evidence of the delivery outcome	Automatic approval of every charge
Freight invoice	Charges for transport and related services	Receipt of payment
Subcontractor bill or settlement	Amount due to a transport supplier	That the customer's invoice has been paid

Terminology varies. “Delivery order” in a port context can authorize release of cargo; it is not automatically the same as a warehouse delivery note. “Bill of lading” is common in sea freight and some other markets, but you should not impose its legal meaning on every Saudi domestic road document. In Saudi road workflows, learn the relevant **وثيقة النقل** and **بيان الحمولة** separately from the customs declaration.

For any document, ask who creates it, who must accept it, which job it refers to and what process it unlocks.

A missing PO number might block customer invoice approval without stopping the truck. Missing release authorization might stop collection entirely. A damaged-goods note might allow physical delivery while opening a claim. These are different exceptions.

8 How a transport company makes money

Revenue has a service basis

A carrier may charge per trip, per pallet, per tonne, per kilometre, per vehicle-day or through a dedicated-capacity contract. A rate can also depend on lane, equipment, urgency and service conditions.

Spot pricing covers individual or short-term requests. **Contract pricing** applies agreed rates or formulas over a period. A rate agreement does not necessarily guarantee volume, and a demand forecast does not necessarily guarantee payment.

Extra charges, often called **accessorials**, can cover waiting, additional stops, special handling or other agreed services. A delay is not automatically billable. The agreement, evidence, approval process and

actual collection matter.

Understand three different cost questions

Incremental cost: What extra cost does this decision cause? This is useful when deciding whether to accept work or use a different assignment.

Allocated full cost: What share of the business's costs should this work carry? This helps evaluate longer-term pricing and profitability.

Cash timing: When must the business pay, and when will it collect? This determines whether it can fund operations.

Fuel and outsourced transport often vary directly with work. Tyre and maintenance wear may be estimated per kilometre, even though the invoice comes later. Salary, vehicle finance, insurance and office costs may continue when a truck is idle. Overtime or additional vehicle rental can change that picture.

A dispatcher saving a journey can reduce fuel and wear. That does not immediately eliminate the vehicle's lease payment. **Contribution toward fixed costs is not the same as net profit.**

Worked example of a return load

Assume a fictional outbound trip is 450 km. The vehicle must eventually return to its starting base. All figures below exclude tax and are teaching assumptions.

The outbound customer pays SAR 2,600. Estimated distance-related cost is SAR 1.40 per km, including fuel and an allowance for distance-related wear. A trip allowance costs SAR 250.

A confirmed return load would pay SAR 1,500. Its pickup and final positioning add 40 km of empty movement. It also adds SAR 80 of incremental time-related cost. Both plans fit the same available resource period.

Item	Outbound then empty return	Outbound plus confirmed return load
Loaded distance	450 km	900 km
Empty distance	450 km	40 km
Total distance	900 km	940 km
Revenue	SAR 2,600	SAR 4,100
Distance-related cost	SAR 1,260	SAR 1,316
Trip allowance	SAR 250	SAR 250
Additional time-related cost	SAR 0	SAR 80
Contribution before fixed costs	**SAR 1,090**	**SAR 2,454**

The return load increases contribution by **SAR 1,364**:

```
SAR 1,500 extra revenue − SAR 56 extra distance cost − SAR 80 extra time cost.
```

Notice that the better plan drives **more** total kilometres. Its advantage comes from earning money on a movement the vehicle otherwise makes empty.

Now change the situation. Suppose the return load would prevent another feasible job contributing SAR 1,400. Its apparent SAR 1,364 improvement becomes a SAR 36 disadvantage compared with that alternative, assuming other effects are equal. This is **opportunity cost**.

A possible return load is also different from a confirmed one. A dispatcher might decide to wait for work, but its revenue remains uncertain. A forecast can inform that decision; it must not silently become guaranteed income.

Own fleet versus subcontractor

Suppose the same outbound customer pays SAR 2,600 and an approved partner offers the whole service for SAR 1,800. That leaves SAR 800 before the operator's remaining overhead and risks.

The owned-fleet option above contributes SAR 1,090, including its empty return. On those assumptions, using the owned vehicle is better by SAR 290.

But subcontracting might release the owned vehicle for another feasible job contributing SAR 500. The combined contribution would then be SAR 1,300. The comparison changes because you considered the work of the fleet together.

Check the full service scope, resource timeline, endpoint, partner availability, expected reliability and any retained costs before applying this logic. A cheap unconfirmed quote is not equivalent to secured capacity.

Profit and cash can diverge

A profitable job can still create a cash gap. Fuel and some supplier charges may be paid before the customer pays its invoice. Missing documents can extend that gap.

This explains why a carrier can care about faster POD collection and billing even when those features do not reduce driving distance. It also explains why reducing accounting costs on a dashboard is different from improving cash in the bank.

9 A dispatch exercise with two trucks and three jobs

This miniature example uses fictional sites A, B and C. It illustrates resource assignment rather than Saudi journey times or driver-hour rules. Assume the drivers and vehicles are eligible for the stated work, and that each truck represents a confirmed vehicle and driver team.

Resources and confirmed jobs

Resource	Ready at site A	Capability
Truck 1	08:00	Box body; rear loading only
Truck 2	09:30	Curtain-side body; suitable for rear or side loading

Job	Movement	Pickup loading must start	Handling and travel
J1	A to B	08:00–10:00	Either truck; 30 min loading, 60 min travel, 30 min unloading; finish delivery by 12:00
J2	A to C	10:00 exactly	Side loading required; 30 min loading, 45 min travel, 30 min unloading; finish by 12:00
J3	B to A	10:30–11:00	Either truck; 30 min loading, 60 min travel, 30 min unloading; finish by 13:00

Treat these as planning estimates for the exercise. Loading and unloading sites at B are assumed to be the same facility, with no transfer travel. No cargo is shared between jobs.

Build the plan

J2 can only use Truck 2. Protect that assignment first.

Truck 1 can perform J1 early, be available at B, and then take J3 back to A.

Resource	Activity	Time
Truck 1	Load J1 at A	08:00–08:30
Truck 1	Travel to B	08:30–09:30
Truck 1	Unload J1	09:30–10:00
Truck 1	Available buffer at B	10:00–10:30
Truck 1	Load J3 at B	10:30–11:00
Truck 1	Travel to A	11:00–12:00
Truck 1	Unload J3	12:00–12:30
Truck 2	Load J2 at A	10:00–10:30
Truck 2	Travel to C	10:30–11:15
Truck 2	Unload J2	11:15–11:45

Assigning Truck 2 to J1 at 09:30 is individually feasible: it finishes delivery at 11:30. But that leaves nobody able to perform J2 at 10:00. A sequence of individually reasonable decisions can produce an infeasible overall plan.

The exercise also shows why completion times matter. Truck 1 arrives at B at 09:30 but becomes free at 10:00. Omitting unloading would exaggerate its availability.

Introduce a disruption

At 09:45, site B says J1 unloading will now finish at 12:00. J3's latest permitted loading start is 11:00. Truck 1 can no longer perform J3 under the agreed window.

Truck 2 is committed to J2. Even if it finishes J2 as planned at 11:45, it is already too late for J3 before accounting for any travel to B.

Reasonable recovery options include obtaining confirmed suitable partner capacity for J3, negotiating a changed pickup window with the customer, or reviewing a broader set of unstarted assignments if genuinely movable capacity exists.

The dispatcher must record the approved recovery and communicate it. Changing a time in the system does not change the customer's agreement.

What this teaches

The dispatch problem combines **eligibility**, **timing**, **sequencing**, **uncertainty** and **commercial consequences**. The map is only one input.

A useful system should be able to explain: "Truck 2 is reserved for J2 because Truck 1 cannot load that cargo. J3 became infeasible because Truck 1's release at B moved beyond the pickup window."

10 Why plans fail in ordinary operations

Event	Immediate effect	Wider consequence
Cargo is not ready	Loading cannot start	Later delivery and return work may become infeasible
Receiver has no unloading capacity	Truck waits	Vehicle, trailer and driver may remain occupied
Quantity differs from the order	Capacity or handling changes	Another vehicle, price revision or customer approval may be needed
Vehicle breaks down	Resource becomes unavailable	Recovery requires a real replacement and possibly cargo transfer
Driver cannot continue	Vehicle may be usable but the team is not	Relief requires eligibility, travel and a handover
Partner does not confirm	Purchased capacity remains uncertain	The operator must secure an alternative before relying on it
Site rejects goods	Delivery is incomplete	Storage, return instructions, claims and billing need attention
POD is missing	Delivery evidence is incomplete	Customer approval or invoicing can be delayed
New urgent job arrives	New demand competes for capacity	Accepting it may displace more valuable committed work

A recovery action has its own workflow. Moving cargo between trucks needs a location, labour, time, equipment and authorization where required. Swapping trailers requires compatible equipment and physical access. An optimizer cannot solve those problems merely by changing resource IDs.

Some delay is recoverable inside a buffer. Other delay destroys the next connection. This is why average transit time alone is not enough: the consequences depend on the remaining schedule.

Also distinguish **exception detection** from **exception resolution**. Knowing a truck is late is useful. Knowing who must act, which options remain feasible and whether the customer approved the change makes that information operational.

11 Where the operational information lives

A logistics company can have several systems and still rely heavily on human coordination between them.

System	Main purpose	Typical information
Transport management system or TMS	Organize transport work	Jobs, assignments, statuses, rates and transport billing
Enterprise resource planning system or ERP	Manage business transactions and accounts	Customers, sales, purchasing, invoices and accounting
Warehouse management system or WMS	Manage goods inside a warehouse	Inventory, picking, staging, loading and receiving
Telematics or fleet tracking	Report vehicle observations	Positions, trips and available vehicle sensor data
Maintenance system	Manage equipment condition and service	Defects, repairs, inspections and availability
Driver application	Communicate assigned work and capture events	Acknowledgements, arrival, loading, delivery and POD
Customer or supplier portal	Exchange information with another organization	Bookings, status, approvals and documents
Email, chat and spreadsheets	Handle flexible communication and local coordination	Requests, exceptions, explanations and working plans

A TMS may include several of these functions; another company may use separate products. **EDI**, or electronic data interchange, means structured business information exchanged between organizations. It is a communication approach, not one universal logistics database.

Opter's feature overview shows how broad a transport system can become: intake, assignment, multi-leg work, driver communication, POD, pricing, invoicing and supplier settlement. Its own feature list also includes AI-assisted intake and route optimization. This is useful context when judging what a new product must improve.

Order, shipment, leg, trip and stop

These words are often used inconsistently across systems. Agree their meanings before importing data.

For learning, use these working definitions:

- **Order:** the customer's commercial request for transport.
- **Shipment or consignment:** the identified goods being transported together under a particular movement arrangement.
- **Leg:** one portion of that movement, such as terminal to terminal.
- **Trip or vehicle run:** a vehicle's planned movement, potentially carrying several shipments or performing sequential jobs.
- **Stop:** a location visit with an activity, such as pickup, delivery or handover.

In a simple FTL operation, one order may map to one shipment and one loaded leg. In a network operation, that relationship becomes many-to-many. A vehicle's empty repositioning can also matter even when no customer shipment is aboard.

Observations are not the complete operational truth

A GPS point establishes where a device was observed. It does not prove that unloading finished, the driver is available or the customer accepted delivery.

A driver pressing “delivered” is a reported event. Finance may still need evidence or a discrepancy resolved. A customer's email can revise the delivery window without immediately updating the TMS.

For important information, preserve both **when the event happened** and **when the business learned about it**. If a release was recorded at 14:00 but only reported at 15:30, a planning decision made at 14:30 could not have used that knowledge.

For an ML engineer, this is a familiar distinction between the real process and its imperfect observations. It also creates a common evaluation error: comparing a historical dispatcher with an algorithm that has access to information received later.

12 Saudi context to understand

The physical and commercial principles above travel across countries. Saudi operations add particular institutions, operating conditions and customer requirements.

Know the purpose of each platform

Name	What to understand first
TGA and its transport service channels	The Transport General Authority governs relevant transport activities; licensing and operating permissions are distinct from daily dispatch
Naql and Logisti	Names you will encounter around transport services and administration; verify the current service route for the activity concerned
Bayan	Road-freight transport documents and truck manifests, as described by its operator
Wasl	Regulatory connectivity and monitoring of relevant land-transport activities; this is distinct from deciding which customer job a truck should perform
Fasah	A trade single window supporting import, export and customs-related processes

Name	What to understand first
Saber	Product conformity processes within the relevant Saudi standards and technical-regulation scope
ZATCA Fatoora	Electronic invoicing requirements and processes

Elm's transport products page describes Logisti, Wasl and Bayan. Its Bayan product guide connects Bayan registration with Naql. These are service descriptions, not a complete statement of which obligation applies to a particular operator.

Fasah's services, SASO's product conformity service and ZATCA's e-invoicing overview explain three different kinds of work. A customs clearance, a product conformity certificate and a transport invoice are not interchangeable documents.

For a domestic dispatch product, learn which outputs or statuses an operator already obtains from these services. The existence of an official system does not establish that you need to replace it or integrate with it in the first learning exercise.

Access and calendars affect feasibility

Truck access can depend on city, route, time, vehicle category and permission. TGA's Riyadh service announcement through the Saudi Press Agency describes scheduled entry during restricted periods. Treat that as evidence that access must be modeled, not as a current timetable for every city.

Loading and receiving calendars can also differ by customer, site and season. Ask for the real calendar and its exceptions. Do not assume the same Friday schedule, Ramadan hours or appointment policy across all customers.

The same discipline applies to permitted weights, driver working and rest rules, dangerous goods and specialist transport: determine the applicable operation and current requirements before setting a constraint. European or US defaults are not Saudi rules.

Language and addresses require operational precision

You may encounter Arabic and English names for the same customer or site, different transliterations, coordinates and informal site descriptions. “Warehouse 2” only makes sense within a particular organization.

The useful data record joins the customer identity, physical gate, coordinates, access instructions and contact. Language support helps people use that record, but translation alone does not resolve an ambiguous destination.

Port and cross-border work add dependencies

A container physically arriving at port does not necessarily make it ready for truck collection. Customs release, shipping-line release, terminal availability and appointments may be separate dependencies.

The empty container's return is another task with an agreed destination and timing. Container demurrage and detention commonly relate to time inside and outside the terminal beyond agreed free

time; actual tariff definitions and combined arrangements vary. They are distinct from a trucker's waiting charge. See Maersk's explanation.

Cross-border road movements add country-specific permissions, border procedures and uncertain clearance time. Learn domestic dispatch first, then study these dependencies as a separate operating pattern.

Learn what Incoterms actually answer

When goods are traded, Incoterms can allocate responsibilities, costs and the point of risk transfer between buyer and seller. They do not by themselves establish ownership transfer or replace the carrier's transport contract.

The practical question is who must arrange and pay for transport, from which point, under the sale agreement. You do not need to memorize all eleven rules before understanding a dispatcher's day. Use ICC's explanation of Incoterms and commercial contracts when studying international freight.

13 The metrics operators care about

A metric is only useful when the event, period and denominator are clear.

Metric	Working definition	Interpretation to watch
On-time pickup or delivery	Eligible jobs meeting the agreed event and window ÷ eligible jobs	Arrival and completion can give different results; preserve the original promise and approved revisions
Empty distance share	Empty km ÷ total km over a defined resource set and period	Low empty distance is not valuable if the work loses money or breaks commitments
Payload utilization	Cargo weight ÷ usable permitted payload for the relevant leg	This says little about volume or floor-space use
Time utilization	Time in a defined useful activity ÷ defined available time	Driving, loaded time and paid time are different definitions
Dwell time	Departure minus arrival at a site	Separate queueing, loading and unloading to find the cause
Contribution per truck-day	Defined contribution ÷ included truck-days	State which costs and idle days are included
Outsourcing spend	Cost of purchased transport in the period	It can rise for good reasons if it enables profitable extra work
POD turnaround	Required evidence received minus delivery time	Missing evidence should not disappear from the report
Delivery-to-invoice time	Invoice issuance minus delivery time for the defined cohort	Track unbilled jobs separately to avoid hiding the worst delays
Plan change rate	Defined changes after publication ÷ defined published assignments	Some changes prevent failure; fewer changes is not automatically better

Metric	Working definition	Interpretation to watch
ETA error	Difference between predicted and actual arrival	Compare predictions made at the same lead time

There is no single perfect utilization number. A full truck waiting all day may have excellent payload utilization and poor productive use of time. A vehicle deliberately held for a guaranteed urgent service may look idle while fulfilling a valuable contract.

When evaluating software, compare plans using the same known orders, constraints, starting resources and information available at the decision time. Otherwise the claimed improvement can come from an easier scenario rather than a better decision.

14 How to connect the domain to software opportunities

Treat every software idea as a proposed improvement to a specific decision or handover.

Operational problem	Possible software help	What you must establish
Requests are incomplete or repeatedly entered	Capture and review structured orders	The actual formats, missing fields and current correction work
Resource availability is scattered	Shared operational view	Who maintains it and how quickly changes arrive
Assignments overlook downstream effects	Feasible planning and comparison	Whether meaningful alternatives exist and the dispatcher controls them
Delays trigger many manual checks	Impact analysis and recovery options	Which commitments can move and who approves changes
Extra charges lack evidence	Event and document capture	Whether the charges are contractually billable and collectible
Delivered work remains unbilled	Readiness checks and exception queues	The real reasons finance cannot invoice

A company with fixed routes and fixed customers may have limited assignment freedom but substantial documentation pain. A mixed carrier with volatile jobs and flexible resources may have more planning opportunity.

Before calling something an optimization problem, ask: **Which decisions can this customer actually change?** If everything is fixed by contracts, the more useful product might improve visibility, exception handling or commercial administration.

For our planned product, your first domain milestone is to explain why one feasible dispatch plan is better than another and why the operator might reject both. That understanding determines which data and rules the software must represent.

15 English and Arabic vocabulary

These are practical learning equivalents, not authoritative translations of legal terms. Operators may use different local wording.

English	Arabic you may encounter	Meaning
Carrier or haulier	ناقل / شركة نقليات	Business undertaking transport
Shipper or consignor	المرسِل / الشاحن	Party sending the goods
Consignee	المرسَل إليه	Party receiving the goods
Freight broker	وسيط شحن / وسيط نقل	Intermediary arranging transport
Dispatcher	مسؤول الحركة / منسق النقل	Person coordinating transport assignments
Fleet	أسطول	Group of vehicles under an operation
Tractor unit	قاطرة / رأس شاحنة	Powered unit pulling a trailer
Trailer	مقطورة	Cargo-carrying unit towed by a vehicle
Pickup	استلام / تحميل	Collection; clarify whether the event is arrival or loading
Delivery	تسليم	Handover at destination
Loading and unloading	تحميل وتفريغ	Putting goods onto and removing them from transport
Shipment	شحنة	Identified goods being moved
Load	حمولة	Cargo carried, or a job in informal speech
Trip	رحلة	A defined vehicle movement or run
FTL	حمولة شاحنة كاملة	Dedicated full-truckload service
LTL	شحن جزئي / حمولة جزئية	Shared-capacity service
Empty running	سير فارغ / رحلة فارغة	Movement without cargo
Backhaul or return load	حمولة عودة	Revenue work on a return or repositioning flow
Time window	نافذة زمنية / فترة مسموحة	Agreed interval for a defined event
ETA	وقت الوصول المتوقع	Predicted arrival time
POD	إثبات التسليم	Evidence of the delivery outcome
Transport document	وثيقة النقل	Record associated with the carriage
Manifest	بيان الحمولة	List of cargo in a movement
Customs declaration	بيان جمركي	Customs record; distinct from the truck manifest
Subcontractor	ناقل متعاقد / مقاول من الباطن	Supplier performing assigned work

English	Arabic you may encounter	Meaning
Waiting charge	رسوم انتظار	Agreed charge linked to qualifying waiting
Freight charges	أجور النقل	Payment for the transport service

16 A practical two week study plan

You can do the initial learning from documents, demonstrations and exercises. Later, anonymized operating records and asynchronous explanations from operators will help distinguish your assumptions from their actual process.

Session	Study task	What to produce
1	Read Sections 1–3	Explain a carrier, forwarder and 3PL without using their names as definitions
2	Draw one customer's shipment chain	Identify goods, information and money at each handover
3	Study vehicles, drivers, sites and cargo	A resource sheet showing location, ready time and eligibility
4	Read the request-to-payment workflow	A job record with separate operational and financial milestones
5	Examine document purposes	Explain what five different documents prove and what they do not
6	Recalculate the return-load example	A cost comparison with explicit assumptions
7	Solve the two-truck exercise without the answer	A feasible schedule including handling time
8	Add the unloading delay	A recovery plan with the approvals it requires
9	Explore Opter's order and dispatch documentation	Map visible fields to operational decisions rather than copying screens
10	Read the Saudi platform descriptions	Explain the difference between Bayan, Fasah and Fatoora
11	Create ten fictional jobs and five resources	A small planning board with incompatible jobs and missing information
12	Define five metrics on your dataset	State denominators, timestamps and missing-data handling
13	Compare FTL with an LTL network	Identify the extra legs, handling and shared-capacity decisions
14	Explain a carrier's day aloud	Describe why a feasible plan can still be commercially unattractive

Useful evidence to study next

When an operator is willing to share material, ask for a small anonymized example: an incoming request, a dispatch sheet, the relevant rate agreement, one exception, a POD and the resulting invoice. A short recorded walkthrough or written explanation can accompany it.

Use the same job across the documents. That lets you reconstruct the actual handovers rather than collect disconnected screenshots. Customer names, driver identities and commercially sensitive fields can be removed when unnecessary.

Questions worth answering from those examples:

- Who can accept a job or change its promised time?
- When is the cargo genuinely ready?
- Who updates a truck's expected release?
- What makes a particular vehicle or driver unsuitable?
- What does the dispatcher know that the system does not?
- When is partner capacity considered confirmed?
- Which changes require customer approval?
- What blocks invoicing after delivery?
- How does the company define a profitable trip?
- Which decisions are fixed, and which are still open?

Public research can teach you the workflow and help build credible exercises. It cannot establish a particular company's error rates, recoverable savings or willingness to pay. Those require evidence from that operation.

Check your understanding

Try answering before reading the explanations.

1. **A truck is at the customer's warehouse. Can you assign its next job?** Only after establishing its release time, equipment and driver availability, and the travel needed to the next pickup.
2. **A truck is only half full. Was the plan wasteful?** Not necessarily. The service may be dedicated, or the binding constraint may be weight, handling, urgency or compatibility.
3. **A return load pays less than the outbound load. Should it be rejected?** Compare incremental contribution, time, endpoint and displaced alternatives.
4. **The shipment is delivered. Can the carrier invoice?** Check the agreement, required evidence, references, charges and any unresolved exception.
5. **An algorithm reduces distance by 10 percent. Is that a business improvement?** Only after confirming equivalent service, feasibility, real cost impact and no harmful displacement of later work.
6. **The driver disagrees with the plan. Is the plan wrong?** Investigate the missing fact or misunderstanding; neither party's confidence establishes feasibility.

17 Further reading

Use these sources for specific questions. Supplier documentation describes that supplier's workflow; it is not a universal operating standard.

Source	Read it to understand
FIATA on freight forwarding	The forwarder's role across the transport chain
DHL road freight	Groupage, direct transport and specialist service categories
Opter features	The breadth of a transport management system
Opter order status	Separate execution, dispatch and financial readiness
Elm transport products	Saudi transport service infrastructure
Elm Bayan guide	Road transport documents and manifests
TGA Bayan user guide	An Arabic walkthrough of the transport-document workflow
Saudi Press Agency on truck entry	Why city access and booking requirements matter
Fasah services	Trade and clearance roles
SASO conformity service	Product conformity work through Saber
ZATCA e-invoicing	The distinction between a freight bill and compliant invoice processing
ICC on Incoterms and contracts	Buyer and seller obligations, risk and costs
Maersk on container time charges	Container free time, demurrage and detention

Keep a learning notebook with four columns: **term or situation**, **what I think it means**, **an operational example**, and **what would verify it**. Use it to turn each unfamiliar field or exception into a concrete question about the business.